Zhuoyang Chen

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Education

University of Pennsylvania Philadelphia, PA
M.S.E in Robotics Aug 2026 - Present

- Coursework: Machine Learning, Control and Optimization, Machine Perception

University of Michigan Ann Arbor, MI
B.E. in Computer Engineering *Aug 2022 - Aug 2026*

- GPA: 3.85/4.0
- Coursework: Automatic Control, Digital Control System, Computer Vision, Algorithmic Robotics, Reinforcement Learning Theory

Shanghai Jiao Tong University Shanghai, China
B.E. in Mechanical Engineering *Aug 2022 - Aug 2026*

- GPA: 3.63/4.0
- Coursework: Honors Calculus, Linear Algebra, Probabilistic Theory, Dynamics and Vibration, Design and Manufactoring

Publications and Manuscripts

- **Z. Chen**, X. Wang, S. Revzen. "Walking on Rough Terrain with Any Number of Legs"
  Bioinspiration & Biomimetics. 2026. (IF=3.5) https://doi.org/10.1088/1748-3190/ae9261
- **Z. Chen**, J. Hu, Y. Ban. "Learning Contact Residuals for In-Hand Manipulation Transfer"
  CoRL Workshop on Bringing Physics Simulation and World Models Together for Robotic Manipulation, 2026.
- J. Cao*, **Z. Chen***, et al. "Unrolling the Action Manifold: Visuomotor Policy Learning via Recursive Cascades"
  (Under review).

Research Experience

Shanghai Institute for Physical Artificial Intelligence and Robotics (PAIR) *May 2026 – Sep 2026*
Research Intern | Advisor: Dr. Yutong Ban, Dr. Jianshu Hu

- Developed an object-centric neural residual dynamics model for contact-rich in-hand manipulation, using surface point cloud to encode geometry and contacts and predict rotation-equivariant linear and angular momentum corrections.
- Built a Real2Sim2Real framework that learns residual dynamics from paired replay of target trajectories, freezes the residual to augment the simulator and finetunes in-hand manipulation policies for improved sim2real transfer.
- Validated sim-to-sim transfer from MuJoCo/MJWarp to Genesis on Allegro and Wuji hands, and sim-to-real transfer from MuJoCo/MJWarp to a real Wuji Hand2, achieving state-of-the art transfer rate and generalizability across objects, tasks and hand embodiments.

University of Hong Kong *Mar 2026 – Sep 2026*
Research Assistant | Advisor: Dr. Andrew Luo, Jiahang Cao

- Designed and implemented a coarse-to-fine visuomotor policy that generates manipulation trajectories through hierarchical, recursive action instantiation using a shared generator.
- Achieved roughly twice the performance of Diffusion Policy and ACT on RoboTwin2.0, outperforming π_0 and RDT with approximately 19M parameters and an inference latency of approximately 2 ms per waypoint. https://sagecao1125.github.io/RCP-Site/
- Investigated state-of-the-art world models such as Cosmos3's representation for few-shot policy adaptation by fitting a light-weight MLP action head on downstream tasks, and using latent manifold analysis technique such as ridge regression probe and UMAP to analyze vision-action correspondence of the representation.

ROAHM Lab, University of Michigan *May 2025 – Aug 2026*
Research Assistant (UM-SURE) | Advisor: Dr. Ram Vasudevan, Dr. Yue Hu

- Extended Tidybot++ with a ROS 2 and MoveIt 2 stack spanning Gazebo, Isaac Sim, and real hardware, and built an Isaac Lab environment for reinforcement learning.
- Built an imitation-learning data collection and training pipeline. Trained and deployed Diffusion Policy in simulation and on hardware, and fine-tuned and deployed OpenVLA, π_0, and $\pi_{0.5}$ on hardware.
- Contributed to an articulated-object reconstruction project with interactive exploration and a Real2Sim2Real pipeline. Trained RL policies on the reconstructed objects using the platform, and evaluated the object's quality by the sim2real success rate of the trained policy
- Developed a neural base-pose critic to fine-tune a long-horizon navigation model for terminal base poses suited to downstream manipulation.
  https://github.com/roahmlab/tidybot_platform.git

BIRDS Lab, University of Michigan *Nov 2024 – May 2026*
Research Assistant | Advisor: Dr. Shai Revzen

- Proposed and implemented a bio-inspired cascaded, hierarchical finite state machine (FSM) controller for a compliant-joint, modular legged robot, enabling stable walking on rough terrain with any number of legs and minimal sensor input.
- Designed a modular robot with compliant joints and contact sensing in Autodesk Inventor and Onshape.
- Validated the distributed FSM control architecture in MuJoCo simulator and deployed an open-loop version to a physical robot using an embedded ESP32 control system.

Teaching Experience

Intro to Engineering *May 2024 – Aug 2024*
Teaching Assistant, UM-SJTU Joint Institute

Intro to Signals and Systems *Jan 2026 – May 2026*
Grader, University of Michigan

Selected Honors and Awards

University of Michigan Ann Arbor, MI

- 2025-2026 EECS Outstanding Achievement Award (3 recipients per year) *Mar 2026*
- 2025-2026 Summer Undergraduate Research in Engineering (SURE) Stipend *August 2025*
- University Honors, Dean's List *Dec 2024, May 2025, Dec 2025*

Shanghai Jiao Tong University Shanghai, China

- Shanghai Jiao Tong University Excellent Graduate (Top 10%) *Aug 2026 2026*
- Excellent Undergraduate Scholarship (Top 10%) *Dec 2023, Dec 2024*
- John Wu and Jane Sun Sunshine Scholarship (Top 6%) *Dec 2023*
- Second Prize, Shanghai Undergraduate Engineering Practice And Innovation Ability Competition *Oct 2023*

Professional Skills

Programming: C/C++, Python, Pytorch, MATLAB

Robotics/Simulation: ROS2, Isaac Sim/Lab, MuJoCo, Gazebo

Embedded/Hardware Tools: ESP32, STM32 (CubeMX/CubeIDE), Raspberry Pi, Nvidia Jetson; PCB design with KiCad

Mechanical CAD: Autodesk Inventor, SolidWorks, Onshape